

DP-DBBA Evaporation Script

Explanation and Operating Logic

File analysed: 3. DP-DBBA Evaporation_v7 copia.py

Prepared for the ICN2 NPG chamber automation workflow

Purpose of this PDF: This document explains what the script does, how its logic is structured, how the GUI and hardware communication work, what safety protections are active, and how to operate the script in practice.

Section	What it explains
1	High-level purpose and role in the NPG synthesis sequence
2	Startup inputs, thickness calculation and output folder
3	Hardware monitored or controlled by the script
4	Complete phase-by-phase workflow
5	GUI layout and live editable controls
6	QMB, thickness and evaporation target logic
7	Keysight current control, ramp-up modes and PID hold
8	Safety behaviour, abort, watchdogs and failsafes
9	Files saved by the script and operator checklist
10	Troubleshooting and function map

1. Executive summary

This script runs the DP-DBBA evaporation stage of the NPG synthesis sequence. It is not just a plotting script: it also controls the Keysight power supply used for the CK-1 crucible coil, writes the external oven PID setpoint, reads several instruments in parallel, guides the operator through the shutter steps, and stops when the calculated CK-1 evaporation thickness target is reached.

- **Main experimental role:** evaporate DP-DBBA after the Heat up + Calibration stage, using the calibration thickness ratio from script 1.
- **Core automation:** heat the CK-1 evaporator under software current control, wait until temperature and rate are ready, ask the operator to open the shutter, measure relative evaporation thickness, then ask to close the shutter once the target is reached.
- **Normal ending:** it does not ramp the Keysight down. It returns the current to the base current and intentionally leaves the output ON for script 4, NPG Annealings.
- **Abort ending:** if the Abort ramp-down button is used, the current is ramped down safely and the Keysight output is switched OFF.

Important distinction: Normal completion and abort behave differently. Normal completion is a handoff to script 4, so the Keysight is left ON. Abort is a safety/manual stop path, so the Keysight is ramped to zero and switched OFF.

2. Where this script fits in the full workflow

Step	Script / action	Role
1	Heat up + Calibration	Obtains the CK-1 / sample thickness ratio used to calibrate the evaporation geometry.
2	Sputtering-Annealing	Performs sputtering and annealing cycles before molecule evaporation, depending on the chosen workflow.
3	DP-DBBA Evaporation - this script	Uses the ratio from script 1 to calculate the required CK-1 QMB thickness for DP-DBBA evaporation.
4	NPG Annealings	Continues after molecule evaporation and performs the later annealing/ramp-down sequence.

The script is designed as a bridge between the calibration stage and the final annealing stage. That is why it asks for the thickness ratio at startup and why, after a normal finish, it keeps the Keysight at base current rather than turning it off.

3. Startup inputs and target-thickness calculation

At startup the script asks for two values in the terminal:

- Run name: used to name the output folder and saved files.
- Thickness ratio from Heat up + Calibration: used to convert the desired sample-equivalent DP-DBBA amount into the CK-1 QMB target thickness.

```
run_name = ask_nonempty_text("Enter the name for this DP-DBBA evaporation run: ")
input_thickness_ratio = ask_positive_float("Enter the thickness ratio obtained in Heat up + Calibration: ")

DP_DBBA_SAMPLE_EQUIVALENT_THICKNESS_A = 623.13 / 94.39
IDEAL_CK1_EVAPORATION_THICKNESS_A = input_thickness_ratio * DP_DBBA_SAMPLE_EQUIVALENT_THICKNESS_A
REAL_SAMPLE_THICKNESS_A = IDEAL_CK1_EVAPORATION_THICKNESS_A / input_thickness_ratio
TARGET_CK1_THICKNESS_A = IDEAL_CK1_EVAPORATION_THICKNESS_A
```

Quantity	Formula / value	Meaning
DP_DBBA_SAMPLE_EQUIVALENT_THICKNESS_A	$623.13 / 94.39 = 6.60165 \text{ A}$	The desired sample-equivalent DP-DBBA thickness used by the script.

IDEAL_CK1_EVAPORATION_THICKNESS_A	input_thickness_ratio x 6.60165 A	The CK-1 QMB target needed to obtain the desired sample-equivalent amount.
REAL_SAMPLE_THICKNESS_A	IDEAL_CK1 / input_ratio	This simplifies back to 6.60165 A; it is printed and saved as a reference.
TARGET_CK1_THICKNESS_A	IDEAL_CK1_EVAPORATION_THICKNESS_A	The actual finish criterion for the evaporation phase.
OVEN_TARGET_TEMPERATURE_C	200.0 C	The external oven PID setpoint written at script start.

Practical consequence: The script does not stop when the sample QMB reaches 6.60 A. It stops when the CK-1 relative thickness reaches the CK-1 target calculated from the calibration ratio.

4. Hardware interfaces used by the script

The script opens serial connections to all devices listed below. The COM ports and baud rates are hard-coded in the device_info dictionary.

Device key in code	Port	Baud	What the script does
CK-1 evaporator QMB	COM4	115200	Zeroes at startup; reads thickness and rate repeatedly.
Sample QMB	COM16	115200	Zeroes at startup; reads sample-side thickness and rate repeatedly.
XGS600 HFIG pressure	COM6	9600	Reads synthesis chamber pressure using the #0002USYNTH command.
Oven PID temperature	COM9	9600	Writes S1 setpoint to 200 C; reads M1 process temperature.
Keysight power supply	COM17	9600	Configures remote mode, voltage/current limits, OVP/OCV, current setpoint and output state.
Arduino CK-1 crucible temperature	COM3	9600	Reads the CK-1 crucible temperature used for ramp/PID/watchdog decisions.

Before running: All COM ports must match the actual Windows device assignments. If one port is wrong or already in use, the script will likely fail during connection setup before the GUI becomes useful.

5. Output folder and saved information

The script saves data next to the Python file, not necessarily in the terminal working directory. The output folder name is generated from the run name and sanitized to avoid forbidden filename characters.

```
custom_folder_name = f"DP-DBBA Evaporation data {safe_sample_name}"
final_folder_path = os.path.join(script_folder, custom_folder_name)
os.makedirs(final_folder_path, exist_ok=True)
```

Saved item	When / why it is saved
run_parameters.txt	Saved at startup. Contains the run name, input ratio, calculated CK-1 target, oven PID target and initial readiness criteria.

oven_pid_setpoint.txt	Saved after the script writes S1=200 C and verifies the PID readback.
Device .txt files	Saved repeatedly during the run. Each device gets its own text file with timestamp/value series.
Graph-only snapshots .png	Saved at important transitions: heating end, target reached, shutter closed, abort/safety events.
summary .txt files	Saved with phase state, current values, setpoints, relative thickness and target values.
final DP-DBBA evaporation plot.png	Saved at the end with the 3x3 final measurement plots.

6. Complete phase-by-phase workflow

Phase	Entry condition	What happens	Exit condition
HEATING_UP	Initial phase after startup.	Keysight output is enabled and current is ramped under software control. The CK-1 temperature, CK-1 rate, pressure, oven PID and Keysight values are monitored live.	CK-1 temperature $\geq$ live temperature target AND average CK-1 rate $\geq$ live rate target.
WAIT_SHUTTER_OPEN	Heating readiness condition is satisfied.	The script tells the operator to open the shutter. Temperature PID continues; current is not frozen.	Operator clicks Open Shutter or types terminal command o.
EVAPORATION	Open Shutter is confirmed.	QMB evaporation window is reset; CK-1 and sample relative thickness start from 0 A for the evaporation window. The script watches CK-1 relative thickness.	CK-1 relative thickness $\geq$ calculated DP-DBBA CK-1 target.
WAIT_SHUTTER_CLOSE	Target CK-1 relative thickness reached.	The script tells the operator to close the shutter and waits for confirmation.	Operator clicks Close Shutter or types terminal command c.
FINISHED / HANDOFF	Close Shutter confirmed after target.	Keysight is set to base current and intentionally left ON for script 4. Snapshots and summaries are saved.	stop_event is set and the script begins cleanup without zeroing the Keysight.
ABORT_RAMP_DOWN	Abort ramp-down button is clicked.	The Keysight current is decreased by 0.010 A every 15 s until zero, then output is switched OFF.	Ramp-down reaches 0 A or an emergency fallback is used.
SAFETY_STOP	Watchdog or hard protection condition occurs.	The script saves safety information, sets current to zero, switches output OFF and stops.	Manual inspection is required before restarting.

Heating readiness logic: The upper rate band is displayed and saved, but it is not used to block the transition. The script moves forward when the temperature target is reached and the average CK-1 rate is at least the target rate.

7. GUI: plots, editable targets and operator buttons

The GUI is a Matplotlib window with a 3x3 plot area and a right-side control panel. It is not a separate application framework: the controls are Matplotlib TextBox and Button widgets.

GUI area	Purpose
3x3 plots	Live plots for CK-1 thickness/rate, sample thickness/rate, chamber pressure, oven PID temperature, Keysight current/voltage and CK-1 crucible temperature.

Editable targets	Allows changing CK-1 temperature target, CK-1 rate target and PID temperature band while the run is active.
Ramp-up mode controls	Switches between Steps ramp and Slope ramp; also edits steps threshold, step period and early/mid/late slope targets.
Manual Keysight current	Lets the operator set a manual current. Automated Steps/Slope/PID corrections pause, but safety checks and watchdog remain active.
Operator actions	Open Shutter, Close Shutter and Abort ramp-down buttons.
Run status / Last action	Shows current phase, shutter state, target values, current readings and recent user/system action.

7.1 Live editable targets

The GUI only asks the user for the CK-1 rate target. The low and high rate band are derived automatically from the target. This prevents inconsistent target/band combinations.

Editable value	Default	Internal effect
Temperature target	242.0 C	Used by heating readiness, PID control and watchdog limits.
CK-1 rate target	0.40 A/s	Used by heating readiness. Low/high band update automatically around this target.
PID band	0.75 C	Defines the temperature interval around the target where PID holds current instead of correcting aggressively.

8. QMB and evaporation-thickness logic

Both QMBs are zeroed at script start. When Open Shutter is confirmed, the script resets the evaporation measurement window: the QMB plot data are cleared and the raw thickness values at that moment become offsets. This makes the plotted and relative thickness during EVAPORATION start from zero.

- **Raw QMB value:** the cumulative value returned by the QMB since hardware zeroing.
- **Adjusted/plotted value:** raw QMB value minus the offset stored at Open Shutter confirmation.
- **CK-1 relative thickness:** the adjusted CK-1 thickness used as the finish criterion.
- **Sample relative thickness:** displayed and saved as a reference, but not the finish criterion.

```
def reset_evaporation_measurement_window():
    for key in QMBs:
        raw_thickness = raw_qmb_last_values[key]['thickness']
        if raw_thickness is not None:
            qmb_thickness_offsets[key] = raw_thickness
    ...
    process_state['baseline_ck1_thickness'] = 0.0
    process_state['baseline_sample_thickness'] = 0.0
    process_state['plots_reset_for_evaporation'] = True
```

When the script stops evaporating: During EVAPORATION, the script checks `ck1_rel >= EVAPORATION_TARGET_CK1_A`. Once this is true, it requests a graph snapshot and moves to `WAIT_SHUTTER_CLOSE`.

9. Keysight current control and ramp-up logic

The Keysight power supply is used to drive the CK-1 crucible coil. The script sends SCPI commands to configure remote mode, protection thresholds, voltage compliance, current setpoint and output state.

Parameter	Value	Meaning
KEYSIGHT_START_CURRENT_A	0.005 A	Initial commanded current when

		automation starts.
KEYSIGHT_BASE_WORK_CURRENT_A	0.640 A	Base working current and normal handoff current for script 4.
KEYSIGHT_SOFT_WARNING_A	0.670 A	Normal software cap: commands are clamped at or below this.
KEYSIGHT_HARD_STOP_A	0.685 A	Measured-current hard stop.
KEYSIGHT_INSTRUMENT_OCP_A	0.690 A	Keysight hardware OCP latch, above the software hard stop.
KEYSIGHT_VOLTAGE_LIMIT_V	2.30 V	Normal compliance voltage limit.
KEYSIGHT_HARD_STOP_V	2.45 V	Measured-voltage hard stop.
KEYSIGHT_INSTRUMENT_OVP_V	2.50 V	Keysight hardware OVP latch, above the software hard stop.

9.1 Steps ramp-up mode

In Steps ramp mode, the script increases the Keysight current by fixed steps while the CK-1 temperature is below the editable steps threshold. The default step is 0.005 A and the default step period is 15 s.

9.2 Slope ramp-up mode

In Slope ramp mode, the script estimates the CK-1 temperature slope from recent Arduino temperature points and nudges the current up or down to approach an early, mid or late target slope. This is designed to make the temperature ramp more linear, not to make current linear.

9.3 Temperature PID hold

Once the CK-1 temperature reaches the live target band, the script switches to temperature PID behaviour. Inside the band it holds the present current and slowly discharges the integral term. Outside the band it computes a P/I/D current correction, clamped to a maximum step of 0.005 A.

PID parameter	Value	Effect
PID_CONTROL_PERIOD_S	3.0 s	Minimum period between PID decisions.
PID_TEMP_BAND_C	0.75 C	Default no-chase band around the target.
PID_KP_A_PER_C	0.0040	Proportional current correction per degree of error.
PID_KI_A_PER_C_S	0.000030	Integral correction for accumulated temperature error.
PID_KD_A_PER_C_PER_S	0.0080	Derivative correction based on error change.
PID_MAX_STEP_A	0.005 A	Maximum current change per PID action.
PID_INTEGRAL_LIMIT_C_S	250.0	Integral anti-windup limit.

10. Safety behaviour and failsafes

- **Soft current cap:** all normal current commands are clamped to the soft cap.
- **Hard current and voltage stops:** measured current or voltage above hard limits causes the output to stop.
- **Protection latch detection:** the script queries OCP/OVP trip status and detects unexpected output-off states.
- **Voltage compliance guard:** if measured voltage reaches the compliance limit, the script reduces current by one soft step instead of waiting for a hard stop.
- **Temperature watchdog:** independent of the PID; it can force current down or stop the run if CK-1 temperature/sensor behaviour is unsafe.
- **Abort ramp-down:** controlled current decrease to zero before output-off.
- **Python exit failsafe:** if the program exits without normal handoff, it tries to shut down the Keysight safely.

10.1 Temperature watchdog

Condition	Default limit	Action
Soft over-temperature	target + 5 C	Reset PID and force current down by up to 0.010 A.

Hard over-temperature	target + 10 C	Set SAFETY_STOP, save summary/snapshot, set current to zero and switch output OFF.
No Arduino reading after automation start	120 s grace	Hard stop.
Stale Arduino reading	180 s	Hard stop.
Impossible temperature value	< -20 C or > 500 C	Hard stop.
Sudden unphysical jump	> 35 C between readings	Hard stop.

10.2 Abort versus normal completion

Path	What triggers it	Keysight result
Normal completion	Target CK-1 thickness reached and Close Shutter confirmed.	Output intentionally remains ON; current is set to base current for script 4.
Abort ramp-down	Abort ramp-down button clicked.	Current ramps down by 0.010 A every 15 s, then output switches OFF.
Safety stop	Watchdog, hard current/voltage, OCP/OVP or unexpected output-off condition.	Output is stopped; current set to zero.
Terminal q / keyboard interrupt	Manual stop command or Ctrl+C.	Emergency shutdown path; output is not preserved for handoff.

11. Parallel threads and what each one does

The script uses multiple daemon threads so that acquisition, control and GUI refresh can happen at the same time.

Thread function	Role
monitor_qmb	Zeroes both QMBs, then repeatedly reads CK-1 and Sample thickness/rate.
read_pressure	Reads chamber pressure from XGS600 roughly once per second.
read_PID	Reads oven PID process value M1 and stores it for plotting.
read_arduino	Reads CK-1 crucible temperature from the Arduino.
read_powersupply	Reads Keysight voltage/current, checks OCP/OVP/output state and hard limits.
automate_keysight_heating	Main current-control thread: steps/slope ramp, PID hold and manual-current pause handling.
temperature_safety_watchdog	Independent CK-1 temperature safety layer.
process_controller	Main state machine for HEATING_UP, shutter confirmations, EVAPORATION and handoff.
user_command_listener	Terminal shortcut listener: o, c, s, i, h, q.

12. Recommended operator procedure

- Confirm that all instruments are connected on the expected COM ports and are not open in another program.
- Start the script from a terminal and enter a clear run name.
- Enter the thickness ratio obtained from Heat up + Calibration.
- Check that the script confirms the oven PID target of 200 C and opens the GUI.
- Watch HEATING_UP. Adjust live targets only if there is an experimental reason to do so.
- When the GUI title/status changes to OPEN THE SHUTTER, physically open the shutter and click Open Shutter.
- During EVAPORATION, monitor CK-1 relative thickness, pressure, temperature and current.
- When the target is reached, physically close the shutter and click Close Shutter.
- After normal completion, do not switch off the Keysight from this script. Start script 4 for NPG annealings / controlled ramp-down.
- Use Abort ramp-down only when you want this script to end by safely ramping the evaporator current to zero and switching the output OFF.

12.1 Terminal shortcuts

Command	Meaning
o	Confirm shutter open.
c	Confirm shutter closed.
s	Save a manual snapshot and phase summary.
i	Print current DP-DBBA target/thickness values.
h	Print live heating targets.
q	Manual stop / emergency shutdown.

13. Troubleshooting guide

Symptom	Likely cause	What to check
Script fails immediately on startup	A serial port cannot be opened, wrong COM assignment, missing pyserial/matplotlib/colorama, or device disconnected.	Check Device Manager COM numbers, close other software using the port, and install dependencies.
Oven PID setpoint write fails	PID does not answer S1/M1 or uses an unexpected numeric format.	Check COM9, PID address 00, baud 9600, and whether the PID is powered and in remote-compatible state.
Current suddenly reads about 0 A	Keysight output may have turned off or OCP/OVP may have latched.	Read the console message. The script explicitly queries OUTP?, CURR:PROT:TRIP? and VOLT:PROT:TRIP?.
Heating takes too long to reach Open Shutter	CK-1 temperature/rate conditions not satisfied, PID too conservative, voltage compliance guard reducing current, or manual current mode left ON.	Check CK-1 T, average rate, current mode, voltage, pressure and status panel.
Relative thickness looks reset after Open Shutter	This is intentional.	The QMB evaporation window is reset so the evaporation plot starts from 0 A.
Close Shutter clicked before target reached	It records a manual close snapshot but the state machine only finishes after WAIT_SHUTTER_CLOSE.	Wait until target reached message, then close and confirm.
The script leaves Keysight ON at the end	Normal DP-DBBA handoff behaviour.	Start script 4. Use Abort ramp-down only if you want this script to turn the Keysight OFF.

14. Function map for code navigation

Code area	Main functions	What to look for
Startup and parameters	ask_nonempty_text, ask_positive_float, build_run_info_text, save_run_parameters	Run name, ratio input, target calculations and saved run info.
GUI controls	setup_live_target_controls, update_live_plot, update_live_dashboard_text	Right control panel, editable targets, buttons and 3x3 live plots.
Live target logic	set_live_heating_targets, ck1_rate_band_from_target, set_live_ramp_settings	How GUI edits update internal target dictionaries.
QMB logic	monitor_qmb, reset_evaporation_measurement_window, relative_ck1_thickness, relative_sample_thickness	Zeroing, raw/offset thickness and relative values.
Keysight logic	configure_keysight_for_automation, keysight_set_current, nudge_keysight_current, read_powersupply	SCPI setup, current cap, live readback and protection checks.
PID / ramp control	automate_keysight_heating, apply_temperature_pid_control,	Current automation during heating and evaporation.

	apply_temperature_slope_control	
Safety	temperature_safety_watchdog, emergency_keysight_shutdown, rampdown_keysight_output	Watchdog, emergency stop and abort ramp-down.
Process state machine	process_controller, confirm_shutter_open, confirm_shutter_closed, hold_keysight_for_next_script	Phase transitions, shutter confirmations and handoff to script 4.
Main execution	main, _failsafe_on_exit, _sigint_handler	Thread startup, GUI loop, cleanup and exit behaviour.

15. One-page logic summary

START

Ask run name and ratio
Calculate CK-1 target thickness = ratio x (623.13 / 94.39)
Create output folder and save run parameters
Write Oven PID S1 = 200 C and verify readback
Open serial connections and start monitor/control threads

HEATING_UP

Enable Keysight output at 0.005 A
Ramp current by Steps or Slope mode
Near target temperature, use PID hold around live temperature target
Watchdog checks CK-1 temperature independently
When CK-1 T >= target AND average CK-1 rate >= target -> WAIT_SHUTTER_OPEN

WAIT_SHUTTER_OPEN

Operator opens shutter and confirms
Reset QMB evaporation window -> relative thickness starts at 0 A

EVAPORATION

Continue temperature PID/watchdog/current safety
Track CK-1 relative thickness
When CK-1 relative thickness >= calculated target -> WAIT_SHUTTER_CLOSE

WAIT_SHUTTER_CLOSE

Operator closes shutter and confirms
Set Keysight to base current, leave output ON for script 4
Save final snapshots/summaries -> FINISHED

ABORT OR SAFETY STOP

Abort: controlled ramp-down to 0 A, then output OFF
Safety: immediate protective stop and output OFF

Bottom line: This script automates the DP-DBBA evaporation stage while keeping the critical shutter actions manual. Its central calculation is the CK-1 QMB target from the calibration ratio; its central control loop is Keysight current regulation based on CK-1 temperature; and its normal end state is a handoff to script 4 with the Keysight left ON at base current.